

CHROMOSPHERE-CORONA TRANSITION REGION MODELS WITH MAGNETIC FIELD AND FLUID FLOW

R. GRANT ATHAY

High Altitude Observatory, National Center for Atmospheric Research¹

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ABSTRACT

Energy balance models of the chromosphere-corona transition region are computed for a segment of average Sun network using assumed magnetic field geometry. The energy fluxes considered include radiation, conduction, enthalpy, and gravitational potential energy, but do not include mechanical heating. Two classes of models are considered: conduction driven and flow driven. In the former class, the thermal conductive flux vanishes at the base of the transition region, and in the latter class the thermal conductive flux increases with depth or remains constant. The conduction driven models require primarily outflow, whereas the flow driven models require downflow. It appears that the flow-driven models give best overall agreement with observations for $3 \times 10^5 \text{ K} \leq T \leq 1 \times 10^6 \text{ K}$, but that neither class of models is satisfactory for $T \leq 1 \times 10^5 \text{ K}$. It is concluded, however, that the flow-driven models have ample energy supply even in the low transition region, and there is no apparent need for additional energy sources.

Subject headings: Sun: chromosphere — Sun: corona — Sun: magnetic fields

1. INTRODUCTION

Energy balance in the chromosphere-corona transition region is of considerable interest in connection with coronal and chromospheric heating. The inward conductive flux inferred from the observed emission measures of spectral lines formed within the transition region (cf. Athay 1976, pp. 395-405; Gabriel 1976a) appears to represent a major heat loss from the corona and may provide a major heat source for the upper chromosphere. Also it is of interest to know whether the transition region requires additional heating by dissipation of mechanical energy flux from deeper layers or whether it is sufficient to assume that all of the mechanical heating occurs higher in the corona (Jordan 1980).

A major difficulty with any theoretical model of the transition region lies in identifying and properly treating the major energy terms. Clearly, radiation and thermal conduction are important and are included in almost all existing models. Although some progress has been made using spherically symmetric models, it is well known that thermal conduction is strongly influenced by magnetic field geometry. Models such as Gabriel's (1976a) that have included a reasonable magnetic structure represent a major improvement over spherically symmetric models. However, as noted by Pneuman and Kopp (1978), the observed downflow of matter in the transition region probably carries an enthalpy flux in excess of the conductive flux. Their model improperly ignores the influences of conduction and magnetic fields, but it clearly illustrates that static models of the transition region are an inadequate basis for discussions of the energy balance.

Still other possibilities exist as energy sinks or sources. Jordan (1980) has suggested, for example, that some of the downward conductive flux may be converted to the nonthermal motion observed as line broadening in the lower transition region. A further possibility suggested here is that some of the downward energy flux is used to heat the rising material in spicules. The rate of upflow in spicules may be estimated from

$$F_p = n_p v a_s, \quad (1)$$

where F_p is the proton flux, n_p is the proton density, v is the velocity, and a_s is the fractional surface area covered by spicules. We adopt $n_p = 6 \times 10^{10} \text{ cm}^{-3}$, $v = 2 \times 10^6 \text{ cm s}^{-1}$, and $a_s = 10^{-2}$ (Beckers 1972), which yields $F_p = 1.2 \times 10^{15} \text{ protons cm}^{-2} \text{ s}^{-1}$. The energy flux, F_H , required to heat this matter to 10^6 K is given by

$$F_H = 5F_p kT = 8 \times 10^5 \text{ ergs cm}^{-2} \text{ s}^{-1}. \quad (2)$$

This is comparable to the downward energy flux in current transition region models.

In this paper we investigate models involving radiation, conduction, and fluid flow constrained by a magnetic field structure similar to that adopted by Gabriel (1976a). Two classes of models are considered: "conduction-driven" models and "flow-driven" models. In the former, the thermal conduction flux decreases downward, and the energy deposited by the conduction is balanced by radiation and an upflow that carries enthalpy and gravitational energy out of the transition region. In the flow-driven models, enthalpy and gravitational energy deposited in the transition region by a downflow are balanced by radiation and a downward increase in conduction flux. No allowance is made for mechanical heating or additional energy sinks. Such

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models still have major limitations, but they are useful for identifying remaining problem areas.

Gabriel (1976*b*) has reported a model of the transition region for a coronal hole combining radiation, conduction, and an outflow compatible with the solar wind flow. However, the single model he gives is in conflict with the observations of downflow in the network and is insufficient to indicate the general properties of models with fluid flow. Recently, Wragg and Priest (1980) have considered models of the corona and transition region including conduction, fluid flow, radiation, and mechanical heating for radially diverging magnetic flux tubes. Again, their models do not illustrate the range of possible models for the transition region, and no attempt is made to match observational data.

II. OBSERVATIONAL CONSTRAINTS

Observed energy fluxes in spectral lines formed in the transition region can be used to derive emission measures (EM) as a function of temperature by standard techniques. For our purposes, we adopt Gabriel's (1976*a*) definition of the emission measure modified by the inclusion of the fractional area a ,

$$EM = an_e^2/d \log T/dy. \quad (3)$$

The solid curve in Figure 1 is extracted from his paper and represents the average empirical EM over the solar disk with $a = 1$. The dashed curve is derived from the solid curve and represents the component of the average emission measure that will be computed from the model.

Computations are done for a model in which the network is represented by a set of parallel bands on a plane surface. The bands are defined by the borders of magnetic flux tubes as shown schematically in Figure 2. A more realistic network model is shown in Figure 2 as the square model in which the network encloses the supergranule cell. The square model is used to obtain the fractional area, a , in equation (3).

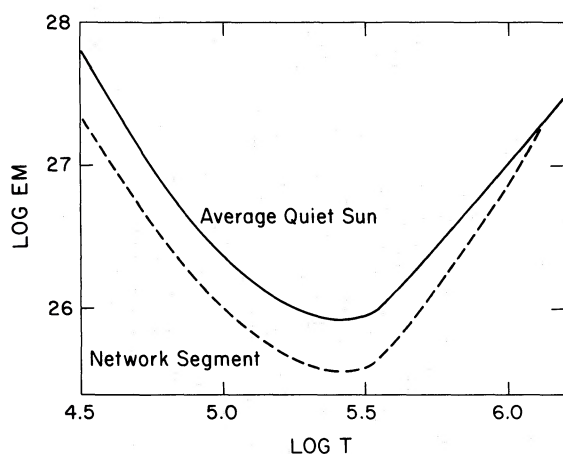


FIG. 1.—Plots of emission measure (see eq. [3]) vs. temperature for the average quiet Sun and a corrected value representing an average segment of network using the square model shown in Fig. 2.

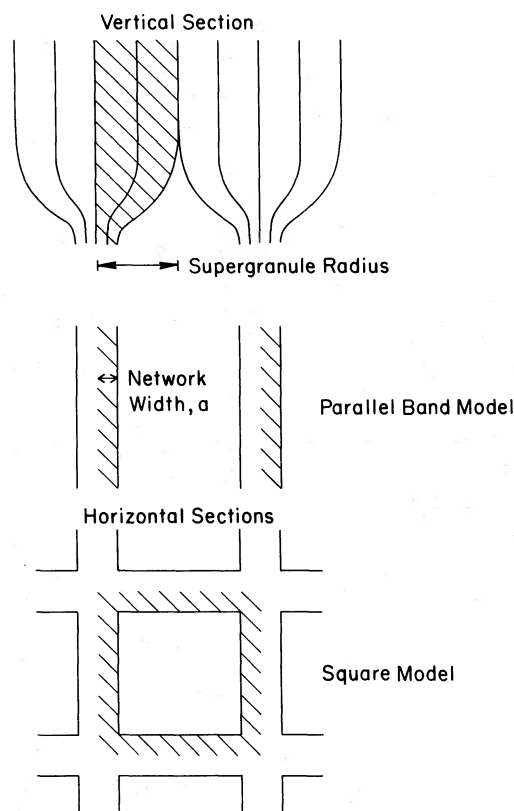


FIG. 2.—A schematic illustration of the assumed magnetic geometry (vertical section) and the network configuration at the base of the temperature scale (horizontal section for parallel band model). A more realistic model of square supergranules with bordering network is shown for comparison. The shaded area in the vertical section shows the region covered by the computations.

We denote the true fraction of the surface area of the Sun covered by network as $A(NW)$ and the fraction covered in the parallel band model as $A(seg)$, and we further denote the true fraction of the radiation flux that arises from the network as $F_R(NW)$. The dashed curve in Figure 1 is then obtained by multiplying the solid curve by $F_R(NW) \times A(seg)/A(NW)$. Using the square model to represent the true Sun, we obtain $A(seg)/A(NW) = (2 - a)^{-1}$, where a is the network width in units of the supergranule radius. Values of a are taken from a representative model in which a varies from 0.27 at the lowest isotherm to near unity at 1×10^6 K. These values vary only slightly for the range of models computed. Values of $F_R(NW)$ are taken from Reeves (1976) and Vernazza and Reeves (1978). They vary from 0.6 at lower temperatures to unity near 1×10^6 K.

An explicit definition of the dashed curve in Figure 1 in terms of model parameters is given in the following section.

The fraction of the solar surface covered by network of relative width a is $a(2 - a)$. Reeves (1976) gives a fractional network area of 0.46, which is relatively independent of temperature for $T \lesssim 3 \times 10^5$. The corresponding

values of a is 0.27. Near 10^6 K the network becomes indistinct. We interpret this as meaning that a increases to unity at 10^6 K and above. We use the value of $a \approx 0.27$ near 10^5 K in the calculations as a boundary condition. Since the models lead naturally to a sudden increase in a to near unity near 10^6 K, it is not necessary to use this as a further constraint.

In addition to the constraints imposed by observed energy fluxes in lines and the width of the network, there is an observed downflow in the network at temperatures near 10^5 K. Lites *et al.* (1975) give an average velocity of 5 km s^{-1} for a Si iv line formed near 8×10^4 K. Doschek, Feldman, and Bohlin (1976) report redshifts for transition region lines of 2 to 15 km s^{-1} relative to chromospheric lines. Brueckner, Bartoe, and VanHosier (1977) show downward velocities ranging up to 20 km s^{-1} . All of these early results are somewhat biased in that they preferentially select the brighter network features where the downflow may be different from the average. Recently, Gebbie *et al.* (1980) have reported an average downflow of 3.9 km s^{-1} in Si iv and 4.2 km s^{-1} in C iv. The C iv lines are formed at 1×10^5 K.

Since the mass upflow in spicules is some two orders of magnitude larger than the average solar wind flux of $4 \times 10^8 \text{ protons cm}^{-2} \text{ s}^{-1}$ at 1 a.u., it is evident that network downflow must balance the spicule upflow (Pneuman and Kopp 1978). Using an upflow of $1.2 \times 10^{15} \text{ protons cm}^{-2} \text{ s}^{-1}$, a network area of 0.46, and a proton density of 6×10^9 at a temperature of 10^5 K, we find an average downflow velocity of 4 km s^{-1} in agreement with the results of Gebbie *et al.* (1980). Rather than impose this downflow on the models, we use it as a desired objective for the models.

The dashed curve $EM(T)$ in Figure 1 together with an average downflow near 4 km s^{-1} at $T = 10^5$ K and $a \approx 1$ at $T = 10^6$ K serve as the desired objectives for the model computations. A value of $a = 0.27$ is imposed near $T = 10^5$ K. Before discussing the computations, however, a few words of caution are appropriate. One is tempted to assume that the value of dT/dy inferred from equation (3) can be used as a measure of the mean vertical temperature gradient in horizontally stratified layers. On the other hand, we know that the transition region is threaded by spicules and loop systems where temperature gradients may be dominantly horizontal. Such structures effectively increase the emitting volume and contribute to the average dT/dy in equation (3). Particularly at low temperatures, the contribution to the EM from spicules and other imbedded features may be very substantial. We have no means of arriving at quantitative estimates of this effect from observations. Thus, we only caution that the empirical EM is an upper limit to what a model with primarily vertical temperature gradient should predict.

A further necessary caution is that it may be somewhat misleading to talk about an average network. Different regions of the network differ in absolute brightness by at least a factor of 10 in transition region lines. Near active regions, the network is brighter and more continuous than for the average Sun. Also, unusually bright regions at 10^6 K do not necessarily overlie unusually bright

regions at 10^5 K. Thus, the gradient of EM varies from point to point as well as the absolute value of EM . Also flow velocities vary greatly from point to point and not necessarily in perfect correlation with the EM . This combination of diversities in the individual features leading to the "average" network could easily lead to an average that is unlike any individual feature. Some evidence in favor of the validity of the average is provided by the essential similarity in the shapes of the emission measure curves for quiet and active regions (Noyes, Withbroe, and Kirshner 1970) as well as for coronal holes (Munro and Withbroe 1976) and even prominences (Orrall and Schmahl 1976). Nevertheless, some caution is necessary in treating the average conditions as representing the average vertical structure.

The observed downflow, of course, is only an average weighted by the EM . In reality, the true net mass flow must be outward and in agreement with the solar wind loss. However, since the solar wind flow is only $\sim 1\%$ of the observed flow, the net downflow is essentially balanced by a corresponding upflow. The observation of a downflow merely indicates that the EM in the downflowing matter exceeds the EM in the upflowing matter. This is not difficult to understand if the matter rises in cool spicules that are mainly below transition region temperatures and descends after being heated to high temperature.

III. MAGNETIC MODEL

For the purposes of the computations in this paper, we utilize the magnetic field only to define the network structure and the flow channels for thermal conduction and fluid motions. Thus, the field geometry is assumed *a priori* and is noninteractive with the plasma flow. This is undoubtedly an oversimplification, but it suffices to illustrate the first-order effects of the magnetic geometry.

To compute the desired magnetic geometry, we assume that the field is potential so that

$$\vec{B} = -\nabla\phi, \quad (4)$$

and

$$\nabla^2\phi = 0. \quad (5)$$

These equations are solved in a two-dimensional box of width unity and height β . Although the width is taken as unity, it represents a supergranule radius of 15,000 km. The field is assumed to be vertical at the two sides and upper boundaries. At the lower boundary the vertical field is specified as a step function given by

$$B_y = B_0, \quad 0 \leq x \leq \alpha,$$

and

$$B_y = 0, \quad x > \alpha. \quad (6)$$

Solutions for B_x and B_y for $B_0 = 1$ are given by

$$B_x(x, y) = \sum_{n=1}^{\infty} C_n \sin n\pi x \sinh n\pi(\beta - y), \quad (7)$$

$$B_y(x, y) = 1 + \sum_{n=1}^{\infty} C_n \cos n\pi x \cosh n\pi(\beta - y), \quad (8)$$

with

$$C_n = \frac{2}{n\pi\alpha} \frac{\sin n\pi\alpha}{\cosh n\pi\beta}. \quad (9)$$

The magnetic field lines are defined by

$$B = \int_0^y B_y dx = \text{constant}.$$

(These solutions were kindly provided by A. Haundhausen.) A set of solutions was carried out for ranges of the parameters α and β to determine their influence on the resultant energy balance models. Figure 3 shows a representative case for $\alpha = 0.1$ and $\beta = 2$ together with a set of isotherms (in units of 10^5 K) to be discussed in § V.

As might be expected from the field configuration shown in Figure 3, the shapes of the field lines between the top of the box and the bottom isotherm, which is fixed in position, are relatively independent of the choice of α and β for $\beta \gtrsim 1.5$ and $\alpha \lesssim 0.3$. Having demonstrated this by numerical computations for several combinations of α and β , we consider in the following only the cases $\beta = 2$ and $\alpha = 0.1$ and 1. The latter corresponds to a uniform vertical field.

Once α and β are selected, the field lines simply define flow tubes of varying diameter. The fluid motion and thermal conduction are assumed to flow along the tubes without additional resistance from the field and to have no component of flow across the field lines, i.e., infinite resistance.

The flow tube along the $x = 0$ axis has a length β , and da/dy is positive along the entire tube. The outer tube at

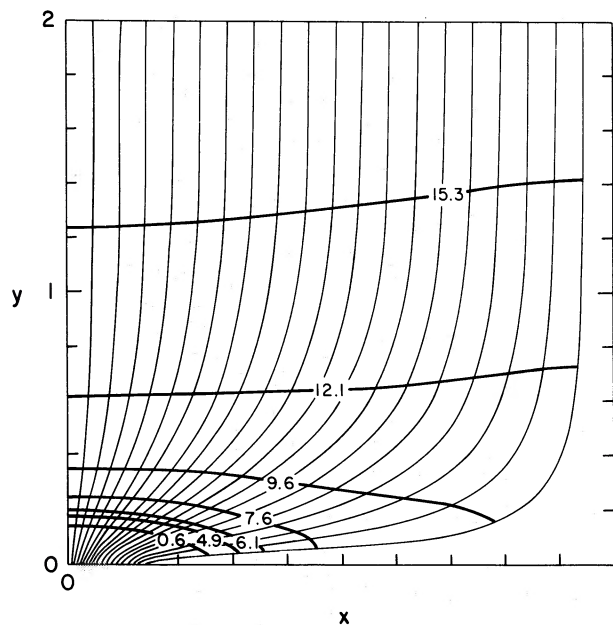


FIG. 3.—Magnetic field pattern from the center of a network segment ($x = 0$) to the center of a supergranule cell ($x = 1$). The magnetic parameters are $\alpha = 0.1$ and $\beta = 2$, and the isotherms are for $n_e T = 6 \times 10^{14}$ ($\delta = 1$), $F_c^1 = 5 \times 10^5$ ergs $\text{cm}^{-2} \text{s}^{-1}$, and $F_c^0 = 0$ ($R = 0$) (see text for explanation).

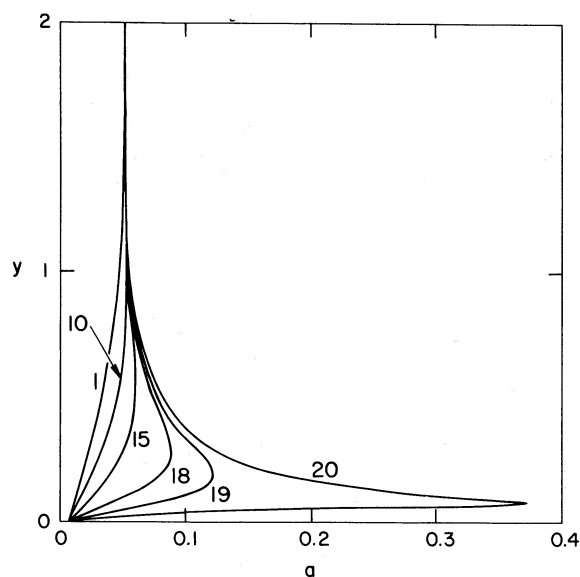


FIG. 4.—Variation of a with y for different flux tubes represented in Fig. 3. The total number of flux tubes is 20, and the higher numbered tubes are near the edge of the network.

$x = 1$ is of length $\beta + 1 - \alpha$ and is much wider than the inner flow tubes. Also, da/dy is negative over much of the tube. This is illustrated in Figure 4, which shows the diameters, a , of selected flow tubes defined by a total of 20 lines of force across the field. The outermost field line (20) has extreme behavior because of the square corner defined by the $y = 0$ and $x = 1$ boundaries. Because of this extreme behavior, we omit the outermost field line in the model computations.

The plot in Figure 4 shows a as a function of Y , which exaggerates the behavior of the outer flow tubes. Since the outer field lines have substantial length below $Y = 0.1$, da/ds , where s is a length measured along the field line, is considerably smaller than da/dy in absolute value. Nevertheless, the markedly different shapes of the flow tubes across the field pattern substantially changes the details of the energy balance across the field.

The only information needed from the magnetic field beyond this point are values of a , da/ds , and dy/ds for the various flow tubes. Thus, as indicated earlier, the magnetic field merely defines the geometry of the tubes in which the energy balance equations are solved.

It is clear, at the outset, that the field geometry shown in Figure 2 is perhaps a gross oversimplification. At coronal temperatures, much of the solar surface has horizontal field components, whereas we have assumed purely vertical fields. Also, it is highly likely that not all of the field lines are confined to the network. Instead the network probably represents a concentration of field lines that make up some fraction of the total field. Introducing such effects would result in changes in a and da/ds . Some insight into the possible effects of such changes can be gained by comparing results of the energy balance models for the inner and outer field lines. Fortunately, these differences are not so great as to destroy all confidence in

the results. This suggests that the effects of a and da/ds are not dominant in the energy balance.

The plot in Figure 3 can be used as a basis for defining a computed quantity that can be compared directly to the dashed curve in Figure 1 representing the network contribution to the observed radiation flux. Specifically, the dashed curve in Figure 1, denoted by EM (network), is defined by

$$EM(\text{network}) = EM^* \int adl = \int an_e^2 \left(\frac{ds}{d \log T} \right) dl, \quad (10)$$

where EM^* is the average EM for the network, dl is an element of length measured along an isotherm, and $\int adl$ is the full length of the isotherm. The lengths are normalized so that $\int adl = 1$ at the top of the y -grid. Note that $\int adl$ is considerably less than unity at lower temperatures. This is consistent with the dashed curve in Figure 1 lying below the average EM for the whole Sun.

IV. ENERGY BALANCE

For the energy balance within each individual flow tube, we assume

$$a \frac{d}{ds} F_R + \frac{d}{ds} [a(F_c + F_{en} + F_g)] = 0, \quad (11)$$

where the flux terms represent, respectively, radiation, conduction, enthalpy, and gravitational potential energy. Although we have formally expressed the radiation loss as a flux divergence, it is more properly thought of as a local energy loss. We specifically omit the kinetic energy term in order to reduce computational time. It is readily shown that for the velocity magnitude we are considering the kinetic energy term is of minor consequence.

The energy fluxes are given by

$$aF_c = \frac{-1.1 \times 10^{-6}}{\gamma} aT^{5/2} \frac{dT}{ds}, \quad (12)$$

$$aF_{en} = 5an_e kTV, \quad (13)$$

$$\frac{d}{ds} aF_g = \frac{d}{ds} (an_e m_H g y V_y), \quad (14)$$

and

$$a \frac{dF_R}{ds} = \gamma a Q_{ne}^2 f(T), \quad (15)$$

where Q is a constant and $f(T)$ is determined from radiation loss functions. Equation (15) serves as a definition of F_R for subsequent discussion. V is measured positively in the downward direction, and s increases downward, i.e., dT/ds is negative. The distance parameter, s , is normalized to unit length, and γs represents the length of the flux tube in units of y . We assume mass continuity within a flow tube so that

$$an_e V = M_0 = \text{constant}. \quad (16)$$

We further assume that

$$n_e T = P_0 = \text{constant}. \quad (17)$$

This, again, is an expedient to reduce computing time. Departures from $P_0 = \text{constant}$ in Gabriel's (1976a) models, which are in hydrostatic equilibrium, are notable only for T considerably above 10^6 K.

With assumption (16), equations (13) and (14) become

$$aF_{en} = 5M_0 kT, \quad (18)$$

and

$$\frac{d}{ds} aF_g = m_H g M_0 \frac{d}{ds} \left(y \frac{\beta dy}{\gamma ds} \right), \quad (19)$$

where we have substituted $\beta/\gamma(dy/ds)$ for V_y/V . Substituting equations (12), (15), (18), and (19) into equation (11) and scaling the parameters to units of 10^5 for T , km s^{-1} for V , and $1.5 \times 10^9 L$ cm for s , we obtain the differential equation

$$\begin{aligned} \frac{d^2 T}{ds^2} + \frac{2.5}{T} \left(\frac{dT}{ds} \right)^2 + \left(\frac{d \ln a}{ds} - \frac{178 L V_0 a_0 \delta}{a T_0 T^{5/2}} \right) \frac{dT}{ds} \\ + 177 \frac{V_0 a_0 \delta}{a T_0 T^{5/2}} \left[\left(\frac{dy}{ds} \right)^2 + y \frac{d^2 y}{ds^2} \right] \\ - 2.32 \times 10^4 \frac{\delta^2 L^2}{T^{9/2}} f(T) = 0. \end{aligned} \quad (20)$$

The term with coefficient 178 is the enthalpy term and that with coefficient 177 is the gravity term. Along with the numerical values of the physical constants, k , m_H , and g , we have used $P_0 = 6 \times 10^{14} \delta$ and $Q = 10^{-22}$. To obtain $f(T)$, we fit the radiation loss function of Summer and McWhirter (1979) with a sum of four Gaussian terms to obtain

$$\begin{aligned} f(T) = 2.5 \exp - 30(0.7 + \log T)^2 \\ + 4.5 \exp - 16(0.1 + \log T)^2 \\ + 4.5 \exp - 16(0.35 - \log T)^2 \\ + 2 \exp - 4(1.1 - \log T)^2, \end{aligned} \quad (21)$$

where T is in units of 10^5 K. The fit is quite good from $T = 0.1$ to $T = 20$.

To solve equation (20), we specify the location of the bottom isotherm (arbitrarily chosen at $T = 0.6$) together with the conductive fluxes at the bottom isotherm and the top of the grid, $y = \beta$. This completely determines the intervening values of F_c , F_R , F_{en} , and F_g . The fixed location of the bottom isotherm is arbitrary, of course, except that the width of the network is known to be ~ 0.27 . All of the computations are done with the bottom isotherm at $s = 0.93$ on each field line. For $\alpha = 0.1$, this gives $a = 0.27$ as required.

Before discussing specific solutions, it is of interest to consider the general ability of equation (20) to account for the observed emission measure. To do this we omit the gravity term, which is of secondary importance, and replace all terms in dT/ds by substituting the EM from

equation (3). For the EM in units of 10^{26} , equation (20) then becomes

$$\left[\frac{d \ln EM}{d \ln T} - 2 \frac{d \ln a}{d \ln T} - \frac{3}{2} \right] + \frac{0.14 a_0 V_0 EM}{a^2 T^{3/2}} - \frac{0.015 f(T) EM^2}{a^2 T^{3/2}} = 0. \quad (22)$$

One feature of equation (22) is of immediate interest. The two terms $d \ln EM/d \ln T$ and $-3/2$ tend to be relatively large at high temperature. Thus, solutions to equation (20) will have a strong tendency for the terms enclosed in brackets to partially cancel, i.e., for $d \ln EM/d \ln T > 0$. This is consistent with the right-hand branch of the curves in Figure 1. On the other hand, there is little possibility of obtaining solutions for which $d \ln EM/d \ln T$ is strongly negative as on the left-hand branch of the curves in Figure 1. If, for example, $d \ln EM/d \ln T$ were negative, solutions could occur only for $V_0 > 0$ (inflow) since all other terms in equation (22) would be negative. Also, using values of EM/a from Figure 1 at $T = 0.5$, the last two terms in equation (22) are of order $4a_0/aV_0$ and -10.7 . Thus, at $T = 0.5$ the radiation term is large and increasing rapidly toward lower temperature. This would require the enthalpy term to increase toward lower temperature, which is inconsistent with equation (18). Thus, at low temperatures the only possible solutions are those for which the radiation (hence EM) and enthalpy terms remain small.

Stated in other terms, the left-hand branch of the emission measure curve would require $d \ln aF_c/d \ln T$ to be approximately constant at a value near $+4$. However, equation (18) gives $d \ln aF_{en}/d \ln T = 1$, for $V > 0$, and numerical computations of F_R show that $d \ln F_R/d \ln T$ is of order unity or less. Furthermore, observations of the network at these temperatures (Reeves 1976) clearly indicate that a is a very slow function of temperature. Thus, there is no physical mechanism in the model capable of maintaining $d \ln aF_c/d \ln T$ at a value near $+4$. It is clear from the outset, therefore, that the models under consideration have no hope of fitting the left-hand branch of the EM curve. We will return to this point in the discussion.

Also, it should be noted that when the enthalpy term is omitted there will be a strong tendency for $d \ln EM/d \ln T$ to be close to $3/2$. This tendency has been noted by several authors discussing models without enthalpy flow.

Some of the important characteristics of the type of models being considered can be deduced without solving equation (20) in detail. To illustrate this, we consider only the fluxes at the upper and lower boundaries. We designate the upper boundary as 1 and the lower boundary as 0. The overall energy balance requires that

$$a_1(F_{en}^1 + F_g^1 + F_c^1 - F_R^1) = a_0(F_{en}^0 + F_g^0 + F_c^0 - F_R^0). \quad (23)$$

To a sufficiently good approximation, we may set $a_0 F_{en}^0 \ll a_1 F_{en}^1$ and $a_0 F_R^0 \ll a_1 F_R^1$. Also, we set $a_1 F_g^1 - a_0 F_g^0 = M_0 m_H g \delta Y$, and we adopt $a_1 = 1$, which gives $M_0 = P_0 V_1/T_1$. This gives

$$V_1 \left(\frac{m_H g y_1}{T_1} P_0 \delta + 5k P_0 \delta \right) = F_R^1 + F_c^1 (R - 1), \quad (24)$$

where $R = F_c^0/F_c^1$. Inserting numerical values for the constants and $y_1 = 3 \times 10^9$ cm, we obtain

$$V_1 = \frac{2.4[F_R^1 + F_c^1(R - 1)]}{(1 + 2/T_1)\delta}. \quad (25)$$

The gravity term is small, and we replace it with an average value obtained by setting $T_1 = 15$. Also, we let $F_R^1/F_c^1 = I\delta^2$ and obtain

$$V_1 = 2.1 F_c^1 \left(I\delta + \frac{R - 1}{\delta} \right). \quad (26)$$

To estimate F_R^1 and F_c^1 , we note from equations (3) and (12) that

$$F_c = \frac{2.5 \times 10^{-6} a T^{3/2} P_0^2}{EM}, \quad (27)$$

and from equations (3) and (15) that

$$F_R^1 = Q \int_{T_0}^{T_1} EM f(T) d \log T. \quad (28)$$

Using observed values of EM and $T_1 = 16$, we find $F_c^1 = 6 \times 10^5$ and $F_R^1 = 3 \times 10^5$ ergs $\text{cm}^{-2} \text{s}^{-1}$. Thus, from the empirical model $I\delta^2 \approx 0.5$.

Equation (26) suggests two general classes of models: one class for which $R \ll 1$ and a second class for which $R \gtrsim 1$. The former class is conduction driven and has $V_1 < 0$, i.e., outflow. The latter class is flow driven and has $V_1 > 0$ as observed. In the conduction driven models the flux F_c^1 is dissipated within the transition region, whereas in the flow-driven models $a_0 F_c^0 > a_1 F_c^1$ which requires that the downward flux $F_{en}^1 + F_g^1$ produce both the radiation loss and the increase in aF_c within the transition region.

Note that equation (26) is independent of a and, hence, independent of the magnetic geometry so long as the fluxes in equation (23) are the total fluxes at the top and bottom of the transition region. Also, equation (26) is valid for the individual flux tubes as well as for the integrated network. Note, also, that for the outer flux tubes for which I is increased because of the larger value of F_R from these tubes, there will be a stronger positive component to V_1 . For the $R = 0$ models this will mean decreased outflow in the outer flux tubes, and for the $R \geq 1$ models it will mean increased downflow in the outer tubes. Unfortunately, no observations of the flow velocities as a function of position within the network have been reported.

V. CONDUCTION-DRIVEN MODELS, $R = 0$

Even though it is clear from equation (26) that this class of models will produce an outflow, such models are still of

interest. A set of models of this class computed for ranges of δ and F_c^1 are shown in Figure 5. The values of EM are integrated along the length of the isotherms as defined by equation (10). Isotherms for the case $\delta = 1$ and $F_c^1 = 5 \times 10^5$ are shown in Figure 3. For $\delta = 0.5$ and 2 only the results for one value of F_c are shown in Figure 5. The results for EM for the other values of F_c have essentially the same relative values as for $\delta = 1$. Changes in F_c are equivalent to changes in dT/ds which influence both the conduction and enthalpy terms, and, in addition, change the EM . Changes in δ change the radiation, gravity, and enthalpy terms without directly changing the conduction terms.

Average values of V at the top of the grid ($y = \beta$) for the models of this class are shown in Figure 6. Inflow occurs only for low values of F_c^1 and large values of δ . This is as expected since F_R increases in proportion to δ^2 , and when $F_R > F_c^1$ inflow provides the energy deficit. The exact velocities shown in Figure 6 are in relatively close agreement with those predicted from equation (26).

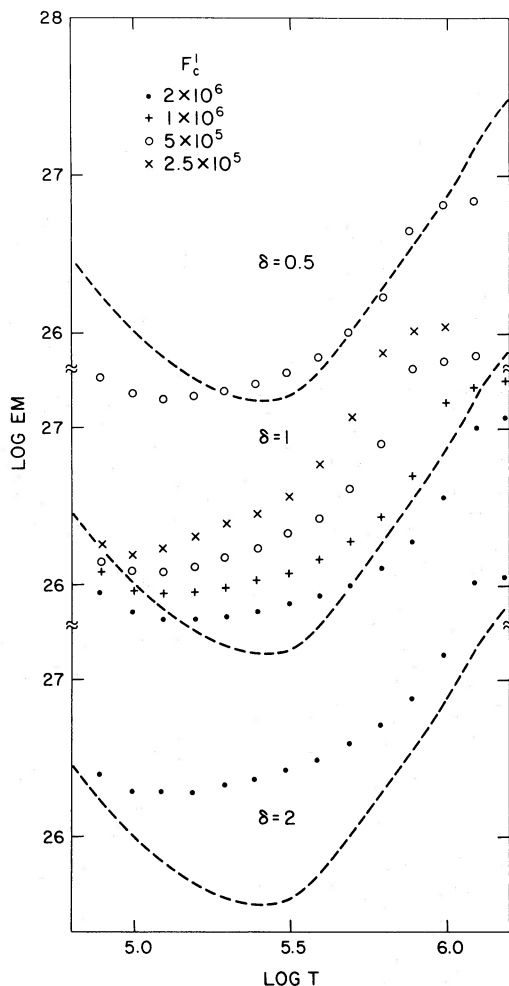


FIG. 5.—Computed emission measures for ranges of δ ($= n_e T/6 \times 10^{14}$) and F_c^1 for the case $F_c^0 = 0$ ($R = 0$). The dashed curves are the same as the dashed curve in Fig. 1.

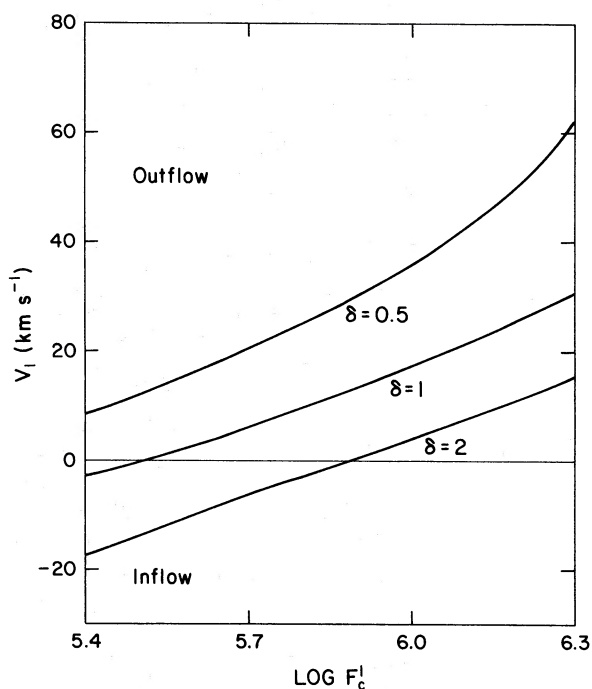


FIG. 6.—Average velocity near the top of the y -grid for ranges of δ ($= n_e T/6 \times 10^{14}$) and F_c^1 for the $F_c^0 = 0$ ($R = 0$) case.

For each of the models shown in Figures 5 and 6 the value of a at $T = 10^5$ K is near 0.3. Values of T_1 at the top of the grid vary from $\sim 2.2 \times 10^6$ K for $F_c^1 = 2 \times 10^6$ to $\sim 1.1 \times 10^6$ K for $F_c^1 = 2.5 \times 10^5$. For the former value the ratio of V at 10^5 K to V at the top of the grid is 0.15 and for the latter it is 0.3.

As noted earlier, values of V for the individual flux tubes vary systematically across the field. The larger flux tubes in the outer portions of the field radiate more than the inner flux tubes because of their increased volume. Thus, for the same value of F_c^1 the outer tubes show less outflow. For $\delta = 1$ and $F_c^1 = 5 \times 10^5$, the outermost tube has a velocity close to zero, whereas the innermost tube has a velocity of -7.4 km s^{-1} .

For this class of models in which F_c is dissipated within the transition region, we have effectively maximized the radiation output. This happens, of course, because it is a decrease in dT/ds that dissipates F_c , and the decrease in dT/ds increases the EM . Most of the dissipation of F_c tends to occur at intermediate temperatures near 3×10^5 K, and it is characteristic of this class of models to produce too large an emission measure in this temperature range.

At temperatures near 10^6 K, several of the models in Figure 5 show a "knee" in the computed EM . This is due mainly to excess emission at the isotherms reaching into the corner of the field where a is unusually large. Aside from the knee, the slopes of the EM curves for the models are consistent with the observational data in the high temperature range.

We noted in § II that the empirical EM curve must be taken as an upper limit to the values predicted by a

satisfactory model because not all of the observed emission is likely to come from layers that are stratified perpendicular to the flux tubes. From this standpoint the plots in Figure 5 give an unsatisfactory fit to the data except in the range near 10^6 K. They are additionally lacking in that they lead to strong outflow.

Models of this class are not improved by adopting less severe magnetic geometry since this only tends to enhance the EM at low temperature relative to that at high temperature.

VI. FLOW-DRIVEN MODELS, $R \geq 1$

There is no conclusive observational evidence that the conduction flux decreases inward in the transition region. In fact, the observed downflow in the network together with the inferred value of F_c^1 are consistent only with the class of models in which the total conductive flux aF_c increases with depth. Since in the network models used here a decreases from 1 at the top of the transition region to ~ 0.3 at the base, F_c must increase by more than a factor of 3.3 from the top of the transition region to the base. In such models, the driving energy is the downflow, and both the increased thermal conduction and radiation loss are the outputs.

Figure 7 shows three models for $R = 1$, $R = 2$, and $R = 3$. The models are computed for $F_c^1 = 5 \times 10^5$ and $\delta = 1.0$. Downflow velocities at the top of the model grid

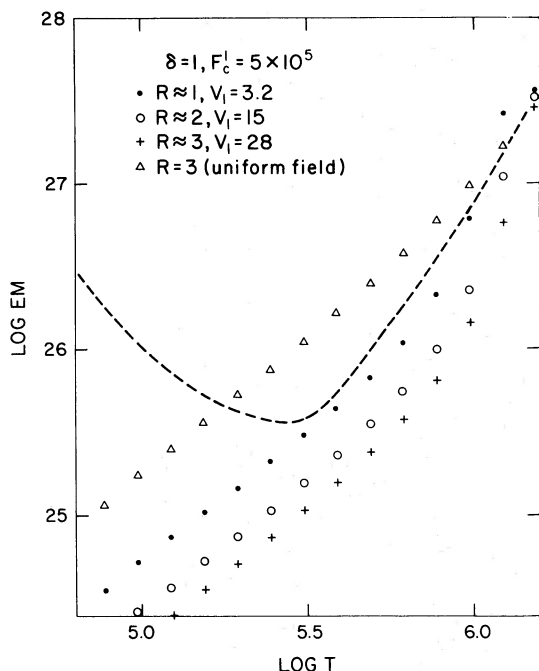


FIG. 7.—Computed emission measures for $n_e T = 6 \times 10^{14}$ ($\delta = 1$), $F_c^1 = 5 \times 10^5$, and $R \approx 1, 2$, and 3 for network models ($R = F_c^0/F_c^1$). A comparison case for a uniform, vertical magnetic field and $R = 3$ is shown also. Values of the average inflow velocities near the top of the y -grid are given in the figure. The dashed curve is the same as the dashed curve in Fig. 1.

are given in the figure. Velocities at the 1×10^5 K level are approximately one-fifth of the values at the top. The downflow is strongest at the outer flux tube and weakest at the inner flux tube.

The models in Figure 7 are more satisfactory than those in Figure 5 both in that they tend to remain below the empirical EM curve and have downflows of the correct order. Perhaps the best overall agreement with both EM and V is for R between 1 and 2. However, it should be noted from equation (26) that the data can be fitted equally as well with other combinations of δ , R , and F_c^1 . The combinations $\delta = 1.5$, $R = 2.12$, and $F_c^1 = 2.2 \times 10^5$ give essentially the same results as $R = 2$ in Figure 7. Ranges of uncertainty in δ and F_c^1 are estimated to be of the order of $0.5 \leq \delta \leq 2$ and $2 \times 10^5 \leq F_c^1 \leq 1 \times 10^6$.

As can be seen from inspection of Figure 7, however, the larger values of R give too low an EM for $T < 1 \times 10^6$ K. This is because the EM is proportional to a/F_c , and we have forced aF_c to increase as a decreases, which requires a rapid increase in F_c . In fact, most of the increase in F_c occurs in layers that overlap the layers where a is decreasing rapidly. This is evidenced by the large negative values of $d \ln EM/d \ln T$ for $T \gtrsim 1 \times 10^6$ K in Figure 7.

Since $aF_{en} \propto T$, most of the enthalpy flux is dissipated at depths where, say, $T \geq \frac{1}{3}T_1$. This means that aF_c must do most of its increasing in this layer and become essentially constant below $T \approx 5 \times 10^5$ K. When this happens, $d \ln EM/d \ln T \approx \frac{3}{2}$, as shown by all the models in Figure 6. Also, absolute values of the EM are proportional to R for fixed values of F_c^1 , as required by equation (27).

Note that the arguments in the preceding leading to the conclusion that $d \ln EM/d \ln T = \frac{3}{2}$ when aF_c is constant are independent of the value of a . This is illustrated by the results in Figure 7 for a case $\alpha = 1$ (uniform field) and $R = 3$. The EM curve is parallel to those for the network models for $T \leq 5 \times 10^5$ K, but it lies above the corresponding curve for the network model by a factor of ~ 10 . This latter occurs because $EM \propto a/F_c$, whereas we have kept aF_c the same in the two models. Thus, the ratio of EM for the two models is proportional to the ratio of a^2 , which is close to 10.

At the top of the y -grid, of course, the uniform field model gives the same results as the network models since even in network models the field is vertical and uniform at the top boundary. Thus, the EM curves for the two magnetic models diverge rapidly as T decreases and become parallel at lower temperatures. The downflow velocity required to drive the vertically uniform model is somewhat larger than in the network model because of the larger radiation flux from the uniform model.

Since the large value of $d \ln EM/d \ln T$ at temperatures just above 1×10^6 K in the network models is due in part to the decrease of a , it is clear that a less severe magnetic geometry would improve the agreement between the computed and empirical values of EM for the larger values of R . This is unlike the situation for the $R = 0$ models where $d \ln EM/d \ln T$ is somewhat too small even in the network models.

VII. DISCUSSION

It seems evident from the results in Figures 5 and 7 that flow-driven models have a much greater chance of providing a satisfactory fit to the observed emission measures than do conduction-driven models. In fact, a compromise magnetic model intermediate to the severe model shown in Figure 3 and the uniform model probably can be found that will give satisfactory agreement with both the observed emission measure for $T \gtrsim 3 \times 10^5$ K and the observed downflow. We conclude, therefore, that the downflow is an essential element of a satisfactory model.

A number of outstanding problems remain. These include the relatively large emission measures observed for $T \lesssim 1 \times 10^5$ K, which are unexplained by either the flow or conduction-driven models, and the large values of conduction flux at the base of the transition region in the flow-driven models. The radiation flux (not including $\text{Ly}\alpha$) arising at temperatures between 3×10^4 K and 3×10^5 K is $\sim 1.5 \times 10^5 \text{ ergs cm}^{-2} \text{ s}^{-1}$. When $\text{Ly}\alpha$ is included, the radiation flux increases to $\sim 4.5 \times 10^5 \text{ ergs cm}^{-2} \text{ s}^{-1}$. The conduction flux at the level where $T = 3 \times 10^5$ K in the flow-driven models is more than sufficient to balance the radiation flux from the lower layers, but in the type of models considered here there appears to be no way to distribute the conduction flux in the manner required by the radiation flux. In view of the simplicity of the models, however, this in no way discredits the concept that the conduction flux is the main energy source for the lower transition region.

At temperatures near 1×10^5 K, spectral line widths indicate nonthermal motions of the order of 20–25 km s^{-1} . Such motions could, and probably do, provide important additional energy fluxes as suggested by Jordan (1980). Upcoming spicule material loses visibility without any widespread evidence of lateral expansion (cf. Beckers 1972). Also, the implied balance between the spicule upflow and the downflow of network material requires that the spicule material be heated to high temperatures. Again, the models contain no appropriate mechanism for such an energy conversion, which, however, is not to be taken as an argument against such a possibility.

A further problem currently under investigation concerns the maximum temperature to which the spicule matter is heated. In this paper, we have assumed that the spicule matter is heated to coronal temperatures, and, thus, that the downflow is continuous throughout the full

range of temperatures considered. In fact, this may not be the case. The network downflow has been reported up to temperatures of 2×10^5 only (Doschek, Feldman, and Bohlin, 1976), and it may be that the downflow is restricted to some maximum temperature well below 10^6 K. Such an effect would reduce the overall importance of the enthalpy term and give a class of models intermediate to those shown in Figures 5 and 7.

In conclusion, there seems to be no compelling reason to seek sources of energy for the transition region other than thermal conduction and the downflow of coronal material. The failure of the models to account for the structure of the lower transition region could be due to the omission of important internal physical processes related to transition region dynamics rather than to external heat sources.

The question of energy balance in coronal holes is of special interest. Munro and Withbroe (1976) give radiation fluxes for coronal holes that are essentially indistinguishable from the average Sun up to $T \approx 1 \times 10^6$ K. Thus, the *EM* curve for coronal hole network is essentially the same as for the average network. On the other hand, the coronal *EM* in coronal holes is sharply reduced from the average *EM*, and both eclipse photos and white light coronagraph data show reduced coronal densities in coronal holes. This has led to coronal hole models in which, compared to the average Sun, P_0 is reduced. Since this would reduce the *EM* in the transition region, the effect of reducing P_0 is compensated in the transition region by reducing dT/ds . Thus, in the conventional transition region model for coronal holes F_c is reduced below its average Sun value in just the proper amount to preserve the value of *EM*. Such models clearly are somewhat suspect.

It is not known exactly how coronal hole geometry extends down to heights where spicules are formed. However, the normal spicule flow in a coronal hole covering just 1% of the solar surface could supply the entire solar wind mass flow. Thus, it appears likely that most of the spicule upflow reappears as downflow in the network even in coronal holes (cf. Withbroe 1977).

In the flow-driven models, the flow energy drives both the thermal conduction and radiation. Thus, if the conduction flux, indeed, is less than average one would expect the enthalpy downflow to be less than average also. Specific models for coronal holes are under consideration for future work.

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R. GRANT ATHAY: High Altitude Observatory, National Center for Atmospheric Research, PO Box 3000, Boulder, CO 80307